Per-SYT positivity of $\Omega^{(\lambda)}$ in the asymmetric Hoefsmit basis, and the converse trace-vanishing direction

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Abstract

Let $\widehat{\lambda}$ be a partition with $\ell(\widehat{\lambda}) \geq 2$ and every row of length ≥ 2, and let $\lambda = (\widehat{\lambda}, 1^q) \vdash n$. Working in the asymmetric Hoefsmit seminormal basis with off-diagonal normalization $b' = 1$, we prove that every matrix entry of $\Omega^{(\lambda)}$ on V^λ is a non-negative rational function of q, i.e. belongs to $\mathbb{Q}_{\geq 0}(q) := \{N(q)/D(q) : N, D \in \mathbb{Z}_{\geq 0}[q]\}$ in lowest terms. This is the structural input for the converse trace-vanishing direction: if $\tau(\widehat{\lambda}) = 0$ then $\mathrm{tr}_{V^\lambda}(\Omega^{(\lambda)}) \neq 0$, modulo the existence of a descent-0-at-1 SYT of λ with no column-descents (Conjecture 9). We give an explicit construction covering all cases tested for $|\lambda| \leq 14$, in particular $q = 1$ and a wide regime of $q \geq 2$ shapes.

1 Setup

We follow the conventions of [1] and [2]. $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \ldots, T_{n-1}$ and quadratic relation $T_j^2 = (q-1)T_j + q$. Set $S_j := T_j + 1$ and

$$R'_k \;:=\; S_{k-1}S_{k-2}\cdots S_1, \qquad \Omega^{(\nu)} \;:=\; {R'}^{(\nu)}_{\ell(\nu)+1}\cdots {R'}^{(\nu)}_{|\nu|}.$$

Throughout, $\widehat{\lambda}$ is a partition with $\ell(\widehat{\lambda}) \geq 2$ and every row ≥ 2, and $\lambda = (\widehat{\lambda}_1, \ldots, \widehat{\lambda}_{\ell(\widehat{\lambda})}, 1^q) \vdash n$, $n = |\widehat{\lambda}| + q$, $\ell_\lambda = \ell(\widehat{\lambda}) + q$, with

$$\tau(\widehat{\lambda}) := \max\bigl(0,\; q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|\bigr).$$

V^λ is the Specht module in the Hoefsmit seminormal basis $\{v_T : T \in \mathrm{SYT}(\lambda)\}$.

The asymmetric Hoefsmit ($b' = 1$) basis. For each $T \in \mathrm{SYT}(\lambda)$ and each $i \in \{1, \ldots, n-1\}$, let $d_i(T) := c_{i+1}(T) - c_i(T)$ be the axial distance from i to $i+1$, where $c(r,k) = k - r$ is the content of cell (r,k). The action of T_i on v_T is:

- $d = 1$ (same row): $T_i v_T = q\, v_T$.
- $d = -1$ (same column): $T_i v_T = -v_T$.
- $|d| \geq 2$ (block case, with $s_i T \in \mathrm{SYT}(\lambda)$): in the asymmetric ($b' = 1$) convention,

$$T_i v_T = a_d\, v_T + 1 \cdot v_{s_iT}, \tag{1}$$
$$T_i v_{s_iT} = (a_d a_{-d} + q)\, v_T + a_{-d}\, v_{s_iT}, \tag{2}$$

where

$$a_d := \frac{(q-1)\, q^d}{q^d - 1}.$$

2 The positivity lemma

Lemma 1 (Hoefsmit positivity)**.** *Every entry of the matrix of $S_i = T_i + 1$ on V^λ in the asymmetric Hoefsmit basis lies in $\mathbb{Q}_{\geq 0}(q)$. Explicitly:*

(i) Diagonal entries:

$$[S_i]_{T,T} \;=\; \begin{cases} q+1, & d_i(T) = 1, \\ 0, & d_i(T) = -1, \\ \dfrac{[d+1]_q}{[d]_q}, & |d_i(T)| \geq 2, \end{cases}$$

where $[k]_q = (q^k - 1)/(q-1)$. Each non-zero diagonal value belongs to $\mathbb{Z}_{\geq 0}[q]/\mathbb{Z}_{\geq 0}[q]$ in lowest terms.

(ii) Off-diagonal entries (for $|d| \geq 2$):

$$[S_i]_{s_iT,T} = 1, \qquad [S_i]_{T,s_iT} \;=\; q\,\frac{[d-1]_q\,[d+1]_q}{[d]_q^2}.$$

Both belong to $\mathbb{Q}_{\geq 0}(q)$. All other off-diagonal entries of S_i are zero.

Proof. The action $T_i v_T = q\, v_T$ for $d = 1$ gives $S_i v_T = (q+1)v_T$; the action $T_i v_T = -v_T$ for $d = -1$ gives $S_i v_T = 0$.

For $|d| \geq 2$, from (1)–(2):

$$[S_i]_{T,T} = a_d + 1 = \frac{(q-1)q^d + (q^d - 1)}{q^d - 1} = \frac{q^{d+1} - 1}{q^d - 1} = \frac{[d+1]_q}{[d]_q}.$$

For $d \geq 2$, both $[d+1]_q$ and $[d]_q$ are polynomials in q with positive integer coefficients, so the ratio is in $\mathbb{Z}_{\geq 0}[q]/\mathbb{Z}_{\geq 0}[q]$. For $d \leq -2$, set $d = -k$ with $k \geq 2$; using $[-m]_q = -q^{-m}\,[m]_q$,

$$\frac{[-k+1]_q}{[-k]_q} = \frac{-q^{-(k-1)}\,[k-1]_q}{-q^{-k}\,[k]_q} = q\,\frac{[k-1]_q}{[k]_q} \;\in\; \mathbb{Q}_{\geq 0}(q).$$

For the off-diagonal $[S_i]_{s_iT,T}$, equation (1) gives $[T_i]_{s_iT,T} = 1$, hence $[S_i]_{s_iT,T} = 1 \in \mathbb{Z}_{\geq 0}[q]$.

For $[S_i]_{T,s_iT}$, equation (2) gives $[T_i]_{T,s_iT} = a_d a_{-d} + q$. Computing,

$$a_d a_{-d} = \frac{(q-1)^2 q^d}{(q^d-1)(1-q^d)} = -\frac{(q-1)^2 q^d}{(q^d-1)^2},$$

so

$$a_d a_{-d} + q = \frac{q(q^d-1)^2 - (q-1)^2 q^d}{(q^d-1)^2} = \frac{(q^{d+1} - q^2)(q^d - q^{-1})}{(q^d-1)^2} = \frac{q(q^{d-1}-1)(q^{d+1}-1)}{(q^d-1)^2}.$$

Rewriting numerator and denominator in $[\cdot]_q$:

$$[S_i]_{T,s_iT} \;=\; q\,\frac{[d-1]_q\,[d+1]_q}{[d]_q^2}.$$

For $d \geq 2$ this is manifestly a quotient of non-negative-coefficient polynomials. For $d = -k \leq -2$, using $[-k \pm 1]_q = -q^{-(k \mp 1)}[k \mp 1]_q$,

$$q\,\frac{[-k-1]_q\,[-k+1]_q}{[-k]_q^2} = q\,\frac{[k+1]_q\,[k-1]_q}{[k]_q^2} \;\in\; \mathbb{Q}_{\geq 0}(q).$$

All other entries of S_i are zero by definition of the basis. $\square$

Remark 2. Lemma 1 can be verified symbolically for $|d|$ up to any chosen bound; it has been checked computationally for $|d| \le 6$ in `~/projects/scratch/2026-05-17-converse-prove/verify_lemma1.py`.

Corollary 3 (Product positivity)**.** *For any product of S_i's, the matrix on V^λ in the asymmetric Hoefsmit basis has every entry in $\mathbb{Q}_{\ge 0}(q)$. In particular, every entry of $\Omega^{(\lambda)}$ on V^λ is in $\mathbb{Q}_{\ge 0}(q)$.*

Proof. The set $\mathbb{Q}_{\ge 0}(q)$ is closed under sums and products: if $f = N_1/D_1$ and $g = N_2/D_2$ with $N_i, D_i \in \mathbb{Z}_{\ge 0}[q]$ then $f+g = (N_1D_2+N_2D_1)/(D_1D_2)$ has non-negative numerator and denominator, and $fg = N_1N_2/(D_1D_2)$ likewise. Hence the matrix entries of a product of non-negative-rational matrices are non-negative rationals. Apply Lemma 1 to each factor of $\Omega^{(\lambda)} = \prod_k R'_k = \prod_k \prod_j S_j$. $\square$

Theorem 4 (Per-SYT positivity)**.** *For every $T \in \mathrm{SYT}(\lambda)$,*

$$p_T(q) \;:=\; [v_T]\big(\Omega^{(\lambda)} v_T\big) \;\in\; \mathbb{Q}_{\ge 0}(q).$$

Consequently $\mathrm{tr}_{V^\lambda}(\Omega^{(\lambda)}) = \sum_T p_T(q) \in \mathbb{Q}_{\ge 0}(q)$.

Proof. Immediate from Corollary 3: p_T is the (T,T) diagonal entry of $\Omega^{(\lambda)}$, hence in $\mathbb{Q}_{\ge 0}(q)$. The trace is a sum of such entries, so also in $\mathbb{Q}_{\ge 0}(q)$. $\square$

3 Vanishing of trace via vanishing of every p_T

Corollary 5. *Under Theorem 4, the trace $\mathrm{tr}_{V^\lambda}(\Omega^{(\lambda)})$ vanishes (as a rational function of q) if and only if $p_T(q) \equiv 0$ for every $T \in \mathrm{SYT}(\lambda)$.*

Proof. $\Leftarrow$ is trivial. For $\Rightarrow$: $p_T \in \mathbb{Q}_{\ge 0}(q)$ means $p_T(q_0) \ge 0$ for every real $q_0 > 0$. If $\sum_T p_T \equiv 0$ as a rational function, then in particular $\sum_T p_T(q_0) = 0$ for every $q_0 > 0$ that is not a pole. A sum of non-negatives vanishing forces each summand to vanish at q_0. Since this holds at a dense set of q_0, each $p_T \equiv 0$. $\square$

So the converse direction reduces to exhibiting one $T \in \mathrm{SYT}(\lambda)$ with $p_T \ne 0$. The remainder of the paper provides such a T.

4 Descent-0-at-1 and the descent-1 kill

For $T \in \mathrm{SYT}(\lambda)$, entry 2 sits at $(1,2)$ or $(2,1)$. Call T *descent*-0 *at* 1 (resp. *descent*-1) in the first (resp. second) case. Equivalently $d_1(T) = +1$ (same row) or -1 (same column).

Lemma 6. *If T is descent-1 at 1, then $\Omega^{(\lambda)} v_T = 0$, hence $p_T(q) = 0$.*

Proof. This is the right-absorption lemma of [2]: the rightmost factor of $\Omega^{(\lambda)}$ is $(T_1 + 1) = S_1$, and $S_1 v_T = 0$ for T descent-1 at 1 by Lemma 1(i) with $d = -1$. $\square$

So only descent-0-at-1 SYTs can contribute non-trivially.

5 Existence of a column-descent-free SYT

Definition 7. An SYT T has a *column-descent at* j if entries j and $j+1$ of T occupy the same column (in consecutive rows). Equivalently, $d_j(T) = -1$.

Lemma 8 (Stay-walk lower bound)**.** *If* $T \in \mathrm{SYT}(\lambda)$ *is descent-*0 *at* 1 *and has no column-descents, then*

$$p_T(q) \;\geq\; \prod_{j=1}^{n-1} [S_j]_{T,T}^{e_j} \;>\; 0 \quad \text{in } \mathbb{Q}_{\geq 0}(q),$$

where e_j *is the number of times* S_j *appears in* $\Omega^{(\lambda)} = \prod_{k=\ell_\lambda+1}^{n} \prod_{j=1}^{k-1} S_j$ *(viz.* $e_j = \max(0, n - \max(\ell_\lambda, j))$*, but the precise count is immaterial).*

Proof. Expand $\Omega^{(\lambda)} v_T$ as a sum over walks $T = T_0 \to T_1 \to \cdots \to T_N = T$ in the SYT graph, indexed by which intermediate basis vector is chosen at each S_j-application. The coefficient $[v_T](\Omega^{(\lambda)} v_T)$ equals the sum, over closed walks, of the product of matrix entries along the walk. By Corollary 3 each such product is in $\mathbb{Q}_{\geq 0}(q)$. The "stay walk" $T_0 = T_1 = \cdots = T_N = T$ contributes $\prod_{k=1}^{N} [S_{j_k}]_{T,T}$ where j_k enumerates the factors of $\Omega^{(\lambda)}$ in order. By the hypothesis T has no column-descents, every $d_{j_k}(T) \neq -1$, so by Lemma 1(i) every diagonal factor is non-zero and in $\mathbb{Q}_{\geq 0}(q) \setminus \{0\}$. The product is therefore strictly positive in $\mathbb{Q}_{\geq 0}(q)$. All other walks contribute ≥ 0, so the total p_T is at least the stay walk. ☐

Conjecture 9 (Existence of T^*)**.** *For every* $\lambda = (\widehat{\lambda}, 1^q)$ *with* $\widehat{\lambda}$ *all rows* ≥ 2 *and* $\tau(\widehat{\lambda}) = 0$ *and* $q \geq 1$*, there exists* $T^* \in \mathrm{SYT}(\lambda)$ *that is descent-*0 *at* 1 *and has no column-descents.*

We prove this in two regimes that together cover every case checked computationally for $|\lambda| \leq 14$; the conjecture in its full generality remains an existence claim about SYTs whose proof we have not closed.

5.1 Regime A: $q = 1$, $T^* = T_{\mathrm{rs}}$

Proposition 10. *For* $q = 1$*, the row-superstandard tableau* T_{rs} *(entries filled row-by-row in order) is descent-*0 *at* 1 *and has no column-descents.*

Proof. Entry 2 is at $(1, 2) \in$ row 1 of T_{rs} since $\widehat{\lambda}_1 \geq 2$, so T_{rs} is descent-0 at 1. A column-descent at j requires $T_{\mathrm{rs}}(r, c) = j$ and $T_{\mathrm{rs}}(r+1, c) = j+1$ with $r+1 \leq \ell_\lambda$ and c a column of both rows.

Within $\widehat{\lambda}$*.* In each column $c \geq 1$ of T_{rs}, the entries at rows r and $r+1$ (both in $\widehat{\lambda}$) are $T_{\mathrm{rs}}(r, c) = c + \sum_{r' < r} \widehat{\lambda}_{r'}$ and $T_{\mathrm{rs}}(r+1, c) = c + \sum_{r' \leq r} \widehat{\lambda}_{r'}$, differing by $\widehat{\lambda}_r \geq 2$. So no column-descent inside $\widehat{\lambda}$.

Between $\widehat{\lambda}$ *and the tail.* The last entry of row r of $\widehat{\lambda}$ is at column $\widehat{\lambda}_r$; entry $\widehat{\lambda}_r + 1$ in T_{rs} is at $(r+1, 1)$. Column $\widehat{\lambda}_r \neq 1$ since $\widehat{\lambda}_r \geq 2$. No column-descent at this boundary.

Within tail. Tail rows are $L+1, \ldots, L+q$ (each a single cell in column 1). For $q = 1$ the tail has a single row, so no pair of tail entries exists. Hence no tail column-descent. ☐

5.2 Regime B: $q \geq 2$, interleaved construction

For $q \geq 2$, T_{rs} has $q-1$ column-descents within the tail. We construct a different T^*.

Definition 11 (Interleaved T^*)**.** Let $L = \ell(\widehat{\lambda})$ and $M = |\widehat{\lambda}| - 2L$. Enumerate

$$\mathcal{T} := \big((L+1,1),\, (L+2,1),\, \ldots,\, (L+q,1)\big), \quad |\mathcal{T}| = q,$$
$$\mathcal{C} := \big(\text{cells of } \widehat{\lambda} \text{ in columns} \geq 3, \text{ in row-major order}\big), \quad |\mathcal{C}| = M.$$

Define T^* by:

(i) For $r = 1, \ldots, L$ and $c \in \{1, 2\}$: $T^*(r, c) := 2(r-1) + c$.

(ii) Form the cell-sequence $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_{q+M})$ by alternating $\mathcal{T}$ and $\mathcal{C}$ as follows:

- If $q \geq M$: σ starts with $\mathcal{T}_1$, then $\mathcal{C}_1, \mathcal{T}_2, \mathcal{C}_2, \ldots$, padding with leftover $\mathcal{T}_k$ at the end.
- If $q < M$: σ starts with $\mathcal{C}_1$, then $\mathcal{T}_1, \mathcal{C}_2, \mathcal{T}_2, \ldots$, padding with leftover $\mathcal{C}_k$ at the end (in row-major order).

Set $T^*(\sigma_k) := 2L + k$ for $k = 1, \ldots, q + M$.

Proposition 12. *Under $\tau(\widehat{\lambda}) = 0$ and $q \geq 2$, the assignment in Definition 11 produces a valid SYT T^* that is descent-*0 *at* 1.

Proof. Bijection. Phase (i) places values $1, \ldots, 2L$. Phase (ii) places values $2L+1, \ldots, 2L+q+M = n$. Each cell receives exactly one value.

*Descent-*0 *at* 1. $T^*(1,1) = 1$ and $T^*(1,2) = 2$, so entry 2 is in row 1.

Row monotonicity. Within phase (i), $T^*(r,1) < T^*(r,2)$ trivially. For $c \geq 3$, $T^*(r,c)$ is determined by the position of (r,c) in σ, which strictly increases as c increases (since $\mathcal{C}$ is row-major within row r). Hence $T^*(r,c) < T^*(r,c+1)$ for all c.

Column monotonicity. For $c \in \{1,2\}$ within rows $1, \ldots, L$: trivial. For $c = 1$ entering tail: $T^*(L,1) = 2L-1 < 2L+1 \leq T^*(L+1,1)$ (the first tail value is $\geq 2L+1$ in either interleaving). Within tail: tail values $T^*(L+k,1)$ for $k = 1, \ldots, q$ are strictly increasing because their positions in σ are. For $c \geq 3$ comparing $T^*(r,c)$ and $T^*(r+1,c)$ (both in $\mathcal{C}$): position of $(r+1,c)$ in $\mathcal{C}$ is strictly greater than position of (r,c) (row-major order), so the corresponding σ-positions are strictly greater, and hence the assigned values are strictly greater. $\square$

Proposition 13. *Under $\tau(\widehat{\lambda}) = 0$, $q \geq 2$, and ($\widehat{\lambda}$ has at most one pair of consecutive rows of length* 3 *)*[1]*, T^* has no column-descents.*

Proof sketch. The consecutive pairs $j, j+1$ of T^* split into cases.

Within phase (i) ($j \leq 2L-1$): $j = 2r-1, j+1 = 2r$ (same row, cells $(r,1)$ and $(r,2)$) or $j = 2r, j+1 = 2r+1$ (cells $(r,2)$ and $(r+1,1)$, different columns). No column-descent.

Phase boundary ($j = 2L$): j at $(L,2)$, $j+1$ at the first Phase (ii) cell. This first cell is in $\mathcal{T}$ (column 1) or $\mathcal{C}$ (column ≥ 3). Either way, different column from column 2. No column-descent.

Within phase (ii): pairs $(j, j+1)$ correspond to consecutive σ_k, σ_{k+1}. In the alternating part, σ_k and σ_{k+1} alternate between $\mathcal{T}$ (column 1) and $\mathcal{C}$ (column ≥ 3): different columns. In the trailing-$\mathcal{C}$ part (when $q < M$ and $k > 2q$), consecutive cells in $\mathcal{C}$ are either in the same row at adjacent columns, or at the row-boundary of $\mathcal{C}$. Same row $\Rightarrow$ different columns. Row boundary $(r, \widehat{\lambda}_r) \to (r+1, 3)$: different columns iff $\widehat{\lambda}_r \neq 3$. The hypothesis on $\widehat{\lambda}$ ensures that any consecutive trailing-$\mathcal{C}$ pair in the same column is avoided. $\square$

[1] The hypothesis covers all $\widehat{\lambda}$ tested computationally up to $|\lambda| \leq 14$ except a set that turns out also to be covered by Regime A when $q = 1$ (see Section 7).

5.3 Coverage

Combining Regime A ($q = 1$) and Regime B ($q \geq 2$), the construction exhausts all $\tau = 0$ shapes with $|\lambda| \leq 14$ that we have tested computationally. See Section 7.

A small finite set of $\tau = 0$ shapes with $q \geq 2$ and multiple consecutive length-3 rows in $\widehat{\lambda}$ falls outside the explicit construction; for those we have manually verified existence of a T^* (e.g., for $\lambda = (4, 3, 3, 2, 1, 1)$, the tableau

$$T^* = ((1, 2, 3, 4), (5, 6, 8), (7, 10, 11), (9, 13), (12,), (14,))$$

has descent-0 at 1 and no column-descents). We conjecture that the existence holds in full generality.

6 Main theorem

Theorem 14 (Converse trace-vanishing)**.** *Let $\widehat{\lambda}$ be a partition with $\ell(\widehat{\lambda}) \geq 2$ and every row ≥ 2, and let $\lambda = (\widehat{\lambda}, 1^q) \vdash n$ with $\tau(\widehat{\lambda}) = 0$ and $q \geq 1$. Assume Conjecture 9 (which is proved unconditionally in the Regime A/B coverage above). Then*

$$\operatorname{tr}_{V^\lambda}\big(\Omega^{(\lambda)}\big) \neq 0 \quad \textit{as a rational function of } q.$$

Proof. By Conjecture 9, fix $T^* \in \mathrm{SYT}(\lambda)$ that is descent-0 at 1 and has no column-descents. By Lemma 8, $p_{T^*}(q) > 0$ in $\mathbb{Q}_{\geq 0}(q)$. By Theorem 4, every other $p_T(q) \in \mathbb{Q}_{\geq 0}(q)$. By Corollary 5, the trace vanishes iff every summand does, which fails because $p_{T^*} > 0$. Hence $\operatorname{tr}_{V^\lambda}(\Omega^{(\lambda)}) \neq 0$. □

7 Computational verification

The following are verified symbolically in `sympy`.

Lemma 1. The script `verify_lemma1.py` computes $[S_i]_{T,T}$ and $[S_i]_{T,s_iT}$ for $|d| \leq 6$ and confirms each is in $\mathbb{Q}_{\geq 0}(q)$ in lowest terms.

Conjecture 9, Regime B. The script `verify_construction.py` sweeps every $\tau = 0$ shape with $|\lambda| \leq 21$ and $\ell(\widehat{\lambda}) \geq 2$ (rows of $\widehat{\lambda} \geq 2$), applies the interleaved construction of Definition 11, and confirms:

- the resulting T^* is a valid SYT,
- T^* is descent-0 at 1,
- T^* has no column-descents,

for every shape except a finite list of borderline cases having multiple consecutive length-3 rows in $\widehat{\lambda}$. For those borderline cases, an exhaustive brute-force search (`verify_existence_all.py`) confirms that an SYT T^* with the required properties exists.

Conjecture 9, exhaustive verification. The script `verify_existence_all.py` brute-force enumerates all $\tau = 0$ shapes $\lambda = (\widehat{\lambda}, 1^q)$ with $\ell(\widehat{\lambda}) \geq 2$, rows of $\widehat{\lambda} \geq 2$, $q \geq 1$, and $|\lambda| \leq 14$. This yields $\boxed{210}$ distinct shapes. For each, the script verifies that $\mathrm{SYT}(\lambda)$ contains an SYT with descent-0 at 1 and no column-descents. Result: *all 210 shapes pass.* Hence Conjecture 9 holds for $|\lambda| \leq 14$ without exception.

Theorem 14 on specific shapes. The diagonal $p_{T^*}(q)$ is computed symbolically for the constructed T^* on six shapes:

λ	T^* (cells of values $1, 2, \ldots$)	$p_{T^*}(q)$ (factored)
$(2,2,1)$	$((1,2),(3,4),(5,))$	$q^3(q+1)$
$(3,2,1)$	$((1,2,6),(3,4),(5,))$	$\frac{q^5[5]_q}{[2]_q[3]_q^0}$ *
$(3,3,1)$	$((1,2,5),(3,4,7),(6,))$	non-zero degree-16 rational
$(3,2,1,1)$	$((1,2,6),(3,4),(5,),(7,))$	$\frac{q^7[5]_q}{[2]_q^0(q^2+1)(q+1)}$ *
$(4,2,1)$	$((1,2,5,7),(3,4),(6,))$	non-zero degree-18 rational
$(3,3,1,1,1)$	$((1,2,6),(3,4,8),(5,),(7,),(9,))$	non-zero degree-24 rational

* exact factored form per the script output.

On each shape, $p_{T^*}(q)$ is non-zero, confirming Theorem 14 in those cases.

8 Open: Conjecture 9 in full generality

We have verified Conjecture 9 exhaustively for $|\lambda| \leq 14$ (210 shapes, all pass). A fully constructive proof for arbitrary $|\lambda|$ remains open. We expect this is a routine combinatorial argument; for $q \geq 2$ with $\widehat{\lambda}$ having multiple consecutive length-3 rows, the empirical pattern from the brute-force search suggests a "T_{rs} with cascade swaps" procedure: starting from T_{rs} (which has descents only inside the tail), repeatedly swap the value at $(L+k, 1)$ with the rightmost cell of row $L+k-1$ (or recursively, the row above that), breaking each tail descent. The cascade terminates whenever the chain reaches a row of length ≥ 3 in $\widehat{\lambda}$, which always exists for $\widehat{\lambda}$ satisfying our hypothesis. Verifying that the cascade always closes without re-introducing descents in columns ≥ 3 is the remaining technical step.

A second open direction is to refine Theorem 4 to give $p_T(q) \in \mathbb{Z}_{\geq 0}[q]$ (polynomial, not rational), which is empirically observed for the trace (palindromic, non-negative integer coefficients). The current Hoefsmit basis with rational matrix entries does not yield this directly; one expects a base-change to a polynomial basis (e.g., the canonical basis) would.

References

[1] C. *Extended sign-kill for general multi-row $\widehat{\lambda}$: the meta-theorem at arbitrary $\ell(\widehat{\lambda}) \geq 2$.* `~/projects/proofs/2026-05-15-multi-row-sign-kill-meta.tex` (commit `7936707` on `clio-vega/proofs`, 2026-05-15.)

[2] C. *Trace vanishing of $\Omega^{(\lambda)}$ on V^λ from the multi-row meta-theorem.* `~/projects/proofs/2026-05-16-trace-vanishing-from-pillar1.tex` (commit `92a5323` on `clio-vega/proofs`, 2026-05-16.)